

## 2.4 - 2.5 Proof Practice

Ten proofs, three sections, and the scaffolding comes off as you go. Every reason you need is in the bank below. Line 1 is always the given, word for word. The last line is always exactly what you were asked to prove.

### REASON BANK — every reason on this page is one of these

Given · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector  
 · Definition of congruent angles  
 Segment Addition Postulate · Angle Addition Postulate  
 Distributive Property · Substitution Property of Equality · Transitive Property of Equality ·  
 Reflexive Property of Equality  
 Addition / Subtraction / Multiplication / Division Property of Equality (two moves on one line get both names)

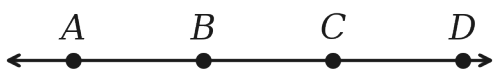
### Part A — The statements are done. Supply each reason.

#### 1. Justify each step (2.4)

Given:  $3(x - 4) = 2x + 1$       Prove:  $x = 13$

| Statements             | Reasons |
|------------------------|---------|
| 1. $3(x - 4) = 2x + 1$ | 1.      |
| 2. $3x - 12 = 2x + 1$  | 2.      |
| 3. $x - 12 = 1$        | 3.      |
| 4. $x = 13$            | 4.      |

#### 2. Overlapping segments (2.4)

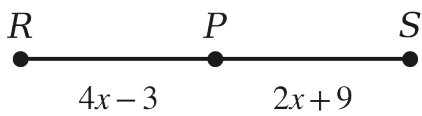


Given: A, B, C, D are collinear in that order, and  $AC = BD$       Prove:  $AB = CD$

*The wholes are given equal. Find the shared piece and cover it.*

| Statements                                           | Reasons |
|------------------------------------------------------|---------|
| 1. A, B, C, D are collinear in that order; $AC = BD$ | 1.      |
| 2. $AB + BC = AC$ ; $BC + CD = BD$                   | 2.      |
| 3. $AB + BC = BC + CD$                               | 3.      |
| 4. $BC = BC$                                         | 4.      |
| 5. $AB = CD$                                         | 5.      |

3. Midpoint (2.5)



Given:  $P$  is the midpoint of segment  $RS$       Prove:  $x = 6$

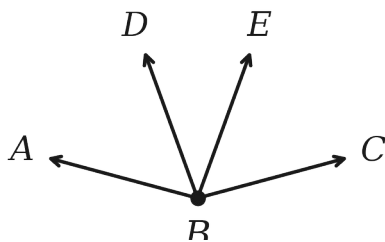
| Statements                             | Reasons |
|----------------------------------------|---------|
| 1. $P$ is the midpoint of segment $RS$ | 1.      |
| 2. $RP = PS$                           | 2.      |
| 3. $4x - 3 = 2x + 9$                   | 3.      |
| 4. $2x = 12$                           | 4.      |
| 5. $x = 6$                             | 5.      |

Part B — The reasons are done. Write each statement.

4. Justify each step (2.4)

Given:  $5(x + 2) = 3x + 22$       Prove:  $x = 6$

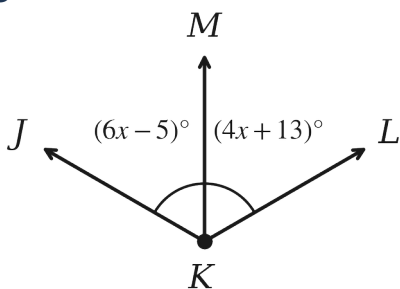
| Statements | Reasons                             |
|------------|-------------------------------------|
| 1.         | 1. Given                            |
| 2.         | 2. Distributive Property            |
| 3.         | 3. Subtraction Property of Equality |
| 4.         | 4. Subtraction Property of Equality |
| 5.         | 5. Division Property of Equality    |

**5. Overlapping angles (2.4)**

Given:  $m\angle ABD = m\angle EBC$       Prove:  $m\angle ABE = m\angle DBC$

*The parts are given equal and you want the wholes. This one runs the ADD direction.*

| Statements | Reasons                              |
|------------|--------------------------------------|
| 1.         | 1. Given                             |
| 2.         | 2. Reflexive Property of Equality    |
| 3.         | 3. Addition Property of Equality     |
| 4.         | 4. Angle Addition Postulate          |
| 5.         | 5. Substitution Property of Equality |

**6. Angle bisector (2.5)**

Given: Ray KM bisects  $\angle JKL$       Prove:  $x = 9$

| Statements | Reasons                              |
|------------|--------------------------------------|
| 1.         | 1. Given                             |
| 2.         | 2. Definition of angle bisector      |
| 3.         | 3. Definition of congruent angles    |
| 4.         | 4. Substitution Property of Equality |
| 5.         | 5. Subtraction & Addition POE        |
| 6.         | 6. Division Property of Equality     |

Part C — On your own

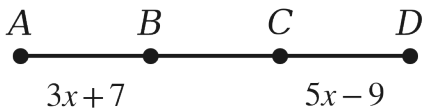
7. Overlapping segments, the other direction (2.4)



Given: W, X, Y, Z are collinear in that order, and  $WX = YZ$       Prove:  $WY = XZ$

| Statements | Reasons |
|------------|---------|
| 1.         | 1.      |
| 2.         | 2.      |
| 3.         | 3.      |
| 4.         | 4.      |
| 5.         | 5.      |

8. Double midpoint (2.5)



Given: B is the midpoint of AC, and C is the midpoint of BD      Prove:  $x = 8$

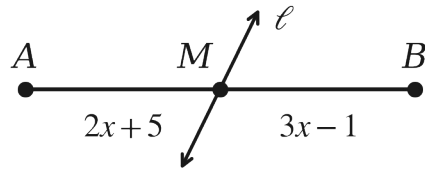
| Statements | Reasons |
|------------|---------|
| 1.         | 1.      |
| 2.         | 2.      |
| 3.         | 3.      |
| 4.         | 4.      |
| 5.         | 5.      |
| 6.         | 6.      |

9. Justify each step — parentheses on both sides (2.4)

Given:  $2(3y + 1) = 4(y + 3)$  Prove:  $y = 5$

| Statements | Reasons |
|------------|---------|
| 1.         | 1.      |
| 2.         | 2.      |
| 3.         | 3.      |
| 4.         | 4.      |
| 5.         | 5.      |

10. Segment bisector (2.5)



Given: Line  $\ell$  bisects segment AB at M Prove:  $x = 6$

*A new word, same move: its definition hands you the two equal halves. Then it is the midpoint proof again.*

| Statements | Reasons |
|------------|---------|
| 1.         | 1.      |
| 2.         | 2.      |
| 3.         | 3.      |
| 4.         | 4.      |

**BEFORE YOU HAND THIS IN**

Line 1 is the given, word for word. The last line is exactly what you were asked to prove. Every reason is a given, a definition, a property, a postulate, or a theorem — and it is in the bank. If a line uses a fact, that fact is written on an earlier line. Quick Check with Substitution: put your  $x$  back into the figure. Both pieces must match.